Frequency Distribution, Cross-Tabulation and Hypothesis Testing

Chapter 20

Matias Ibarburu

UM

UNIVERSIDAD DE MONTEVIDEO

Agenda

Overview
Frequency distribution
A general procedure for hypothesis testing
Cross-tabulation
Statistics associated with cross-tabulation
Hypothesis testing related to differences
Software and practice

Learning objectives

After this chapter you should be able to:

1. Describe the significance of **preliminary data analysis** and the insights it provides.
2. Discuss data analysis associated with **frequencies**: measures of location, variability and shape.
3. Explain **cross-tabulation** and its statistics: chi-square, phi coefficient, contingency coefficient, Cramer's V and lambda.
4. Describe **parametric hypothesis testing** for one sample, two independent samples and paired samples.
5. Recognise the main **non-parametric** counterparts (overview only – non-examinable).
6. Use **Python** (pandas and SciPy) to run frequencies, cross-tabulations and z/t tests.

Why basic data analysis matters

- Once data are prepared (Chapter 19), the researcher runs **basic analyses**: frequency distributions, cross-tabulations and hypothesis tests.
- Many marketing research projects **never go beyond** basic analysis – results are shown as tables and charts.
- Even when multivariate analysis follows, basic analysis:
  - guides the choice of more sophisticated techniques;
  - helps interpret their results;
  - gives quick, low-cost insight that managers can easily understand.
- Running example: **internet usage for personal reasons** – 30 participants (Table 20.1).

Real research: two flavours of basic analysis

Real research: FIBA Brand Tracker (1) – cross-tabulation and chi-square

FIBA commissioned a Brand Health Tracker to see how perceptions of basketball differed across countries.

Interest in basketball	China	Italy	Spain	UK
Avid	187 (31%)	146 (24%)	83 (15%)	85 (14%)
Occasional	115 (19%)	158 (26%)	226 (41%)	219 (36%)
Low	242 (40%)	238 (39%)	187 (34%)	194 (32%)
None	60 (10%)	67 (11%)	55 (10%)	115 (19%)
Total	604	645	609	607

$$\chi^2 = 145.56, \; p \leq 0.0001$$

China and Italy have far more *avid* fans; the UK has the fewest avid fans and the most with no interest. Differences across countries are statistically significant.

Real research: paired t tests

Real research: Catalogues are risky business

Overall perceived risk was compared for 12 product categories bought **by catalogue** vs **in a store**, using one **paired-observation t test** per product (higher = more risk).

Product	Catalogue	Store
Shoes	58.60	50.80*
Pocket calculator	49.62	42.00*
Hi-fi	48.89	41.98*
Digital camera	48.13	39.52*
Sports socks	35.22	30.22*
Perfume	34.85	29.79*

* Significant at the 0.01 level.

The null hypothesis of *no difference in perceived risk* was rejected: catalogue purchases carry significantly more perceived risk.

Running example: internet usage data (Table 20.1)

Table 20.1 Internet usage data

No.	Gender	Famil.	Usage	Att. int.	Att. tech.	Shop	Bank
1	1	7	14	7	6	1	1
2	2	2	2	3	3	2	2
3	2	3	3	4	3	1	2
4	2	3	3	7	5	1	2
5	1	7	13	7	7	1	1
6	2	4	6	5	4	1	2
7	2	2	2	4	5	2	2
8	2	3	6	5	4	2	2
9	2	3	6	6	4	1	2
10	1	9	15	7	6	1	2
11	2	4	3	4	3	2	2
12	2	5	4	6	4	2	2
13	1	6	9	6	5	2	1
14	1	6	8	3	2	2	2
15	1	6	5	5	4	1	2
16	2	4	3	4	3	2	2
17	1	6	9	5	3	1	1
18	1	4	4	5	4	1	2
19	1	7	14	6	6	1	1
20	2	6	6	6	4	2	2
21	1	6	9	4	2	2	2
22	1	5	5	5	4	2	1
23	2	3	2	4	2	2	2
24	1	7	15	6	6	1	1
25	2	6	6	5	3	1	2
26	1	6	13	6	6	1	1
27	2	5	4	5	5	1	1
28	2	4	2	3	2	2	2
29	1	4	4	5	3	1	2
30	1	3	3	7	5	1	2

Gender: 1 = male, 2 = female. Familiarity: 1 = very unfamiliar ... 7 = very familiar (9 = missing). Usage: hours/week. Attitudes: 1 = very unfavourable ... 7 = very favourable. Online shopping/banking: 1 = yes, 2 = no.

Frequency distribution

Many research questions concern a **single variable**:

- How many users of the brand may be characterised as brand loyal?
- What percentage of the market are heavy, medium, light and non-users?
- How familiar are customers with a new product? What is the mean familiarity rating?
- Is the income distribution of brand users skewed towards low incomes?

Frequency distribution

A mathematical distribution whose objective is to obtain a **count** of the responses associated with the different values of **one variable** and to express these counts in **percentage** terms (percentages, valid percentages, cumulative percentages).

Example: familiarity with the internet (Table 20.2)

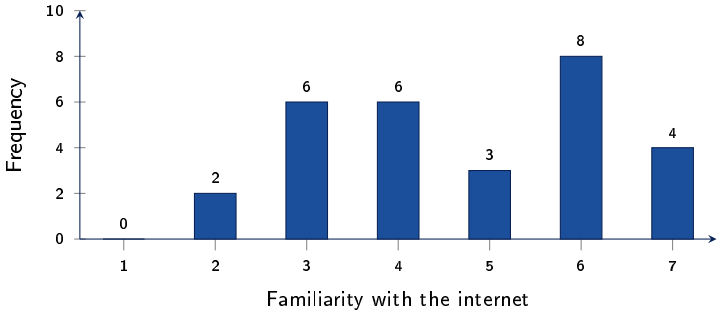
Table 20.2 Frequency distribution of 'Familiarity with the internet'

Value label	Value	Frequency (N)	%	Valid %	Cumulative %
Very unfamiliar	1	0	0.0	0.0	0.0
	2	2	6.7	6.9	6.9
	3	6	20.0	20.7	27.6
	4	6	20.0	20.7	48.3
	5	3	10.0	10.3	58.6
	6	8	26.7	27.6	86.2
Very familiar	7	4	13.3	13.8	100.0
Missing	9	1	3.3		
Total		30	100.0	100.0	

- Reveals **item non-response** (1 of 30) and **illegitimate values** (e.g. 0 or 8 would be errors).
- Detects **outliers** and shows the **shape** of the distribution.
- 58.6% of valid respondents gave a familiarity of 5 or less.

Frequency histogram

Figure 1. Frequency histogram of familiarity with the internet



Values of the variable on the X axis, absolute (or relative) frequencies on the Y axis – compare the observed shape with an expected or assumed distribution.

Real research: frequencies in practice

Real research: FIBA Brand Tracker (2)

Of the 4,288 participants, 900 were *avid* basketball fans. How do avid fans follow basketball? (multiple-response question)

Media used	Frequency	%
Watch dedicated TV	495	55
Read newspaper articles	423	47
Visit websites	315	35
Participate	270	30
Read specialist magazines / blogs	234 each	26
Listen to radio broadcasts	225	25
Attend events	180	20
Receive results via mobile	126	14

FIBA took particular note of **new media** (websites, blogs, mobile) and tracked their growth in later brand-health waves.

Statistics associated with frequency distributions

A frequency table may be too detailed – summarise it with **descriptive statistics**:

Location

- Mean
- Mode
- Median

Variability

- Range
- Interquartile range
- Variance / standard deviation
- Coefficient of variation

Shape

- Skewness
- Kurtosis

Measures of location are also called measures of **central tendency**: adding a constant to every observation shifts the mean, mode and median by that constant.

Measures of location

- **Mean** – the average; appropriate for interval or ratio data:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

For Table 20.2: $\bar{X} = \dfrac{(2 \times 2) + (6 \times 3) + (6 \times 4) + (3 \times 5) + (8 \times 6) + (4 \times 7)}{29} = \dfrac{137}{29} = 4.724$

- **Mode** – the most frequent value (highest peak). Best for categorical data. Here: **6**.
- **Median** – the middle value (50th percentile) when data are ordered. Appropriate for ordinal data. Here: **5**.

Which measure?

Nominal → mode · Ordinal → median · Interval/ratio → mean (uses all the information), but it is sensitive to **outliers** – then report the median as well.

Measures of variability

- **Range** $= X_{\text{largest}} - X_{\text{smallest}}$ (directly affected by outliers). Table 20.2: $7 - 2 = 5$.
- **Interquartile range** – 75th minus 25th percentile (middle 50% of observations). Table 20.2: $6 - 3 = 3$.
- **Variance** – mean squared deviation from the mean; **standard deviation** $= \sqrt{\text{variance}}$:

$$s_x = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{n-1}}$$

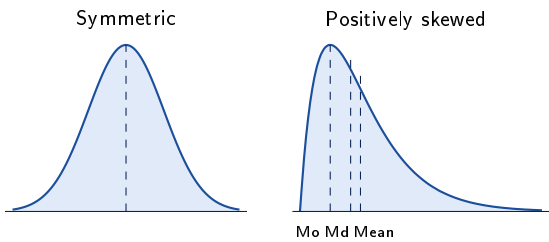
  Divide by $n - 1$: the sample mean makes the sample look less variable than the population. Table 20.2: $s_x^2 = 69.793/28 = 2.493, \quad s_x = 1.579$.
- **Coefficient of variation** $CV = s_x/\bar{X}$ – unitless; meaningful **only for ratio scales** (so not for familiarity).

Managerial relevance: a characteristic with good variability may be a basis for **segmentation**.

Measures of shape

- **Skewness** – tendency of deviations from the mean to be larger in one direction (one tail heavier).
  - Symmetric: mean = median = mode.
  - Table 20.2: skewness = −0.094 (slight negative skew).
- **Kurtosis** – relative peakedness or flatness; 0 for a normal distribution.
  - Positive: more peaked; negative: flatter.
  - Table 20.2: kurtosis = −1.261 (flatter than normal).
- If a distribution is highly skewed or peaked/flat, use procedures that **assume normality** with caution.

Figure 2. Skewness of a distribution



Left: mean = median = mode. Right (positive skew): mode < median < mean.

Hypotheses in marketing research

Basic analysis invariably involves some hypothesis testing. Examples:

- A cinema is attended by more than 10% of the households in a city.
- Heavy and light users of a brand differ in psychographic characteristics.
- One hotel has a more 'luxurious' image than its close competitor.
- Familiarity with a restaurant results in greater preference for it.

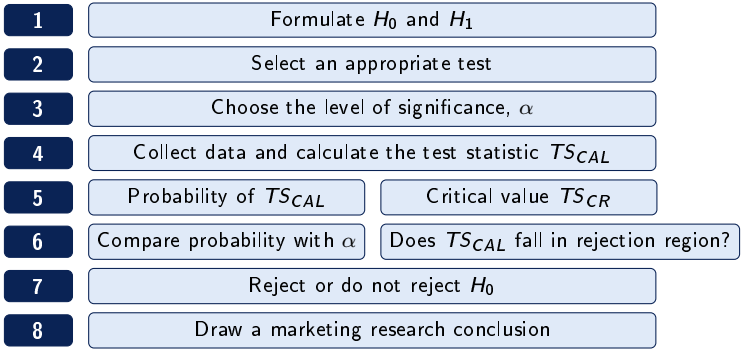
Concepts from Chapter 15 – **sampling distribution**, **standard error** of the mean or proportion, **confidence interval** – are all relevant here.

Running example

A FIBA sponsor will open an **online store** only if **more than 40%** of internet users shop online. In our sample, 17 of 30 users shop online.

The eight steps

Figure 3. A general procedure for hypothesis testing



Step 1: formulate the hypotheses

Null hypothesis H_0

Status quo: **no difference or no effect**. If not rejected, no changes are made. Always the hypothesis that is tested. Refers to a **population parameter** (μ, σ, π), not a sample statistic $(\bar{X}, S, P)$.

Alternative hypothesis H_1

Some difference or effect is expected. Accepting it leads to **changes in opinions or actions**. It represents the conclusion for which evidence is sought.

- H_0 may be **rejected** but never **accepted** on the basis of a single test.
- Formulate H_0 so that its rejection leads to the desired conclusion.

One-tailed test (directional H_1):

$H_0 : \pi \leq 0.40 \qquad H_1 : \pi > 0.40$

Two-tailed test (non-directional H_1):

$H_0 : \pi = 0.40 \qquad H_1 : \pi \neq 0.40$

One-tailed tests are more common in commercial research (there is usually a preferred direction) and are more powerful.

Steps 2 and 3: test statistic and significance level

Step 2. The **test statistic** measures how close the sample has come to H_0; it usually follows a known distribution (normal, t, χ^2). For a proportion:

$$z = \frac{p - \pi}{\sigma_p}, \qquad \sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

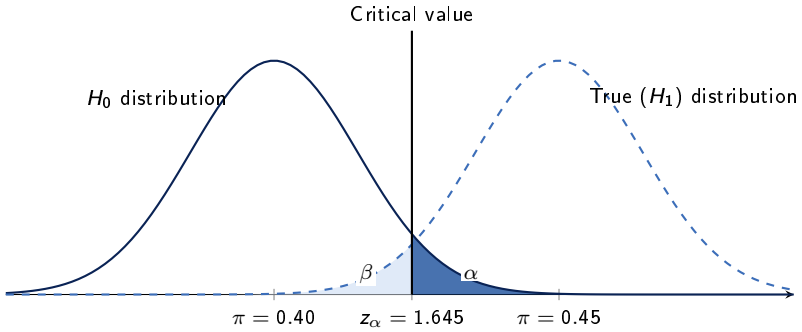
Step 3. Two types of error can occur:

	H_0 true	H_0 false
Reject H_0	**Type I error** (α)	Correct (power $= 1 - \beta$)
Do not reject H_0	Correct	**Type II error** (β)

- α = **level of significance**, set by the researcher (usually 0.05, sometimes 0.01), depending on the cost of a Type I error.
- β depends on the true parameter value; a very small α gives an intolerably large β.
- Increasing the sample size lowers β for a given α, raising the **power** of the test.

Type I and Type II errors

Figure 4. Type I error (α) and Type II error (β) for $H_0 : \pi = 0.40$ vs a true $\pi = 0.45$



Moving the critical value right lowers α but enlarges β – the two errors must be balanced.

Steps 4 and 5: compute the statistic and its probability

Step 4. Sample size reflects desired α, β and budget. 17 of 30 users shop online:

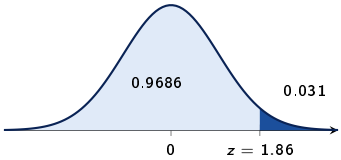
$$p = \frac{17}{30} = 0.567, \qquad \sigma_p = \sqrt{\frac{0.40 \times 0.60}{30}} = 0.0894, \qquad z = \frac{0.567 - 0.40}{0.0894} = 1.86$$

Step 5. The **p value** is the probability of a test statistic as extreme as, or more extreme than, the one observed, **assuming H_0 is true**.

- Area to the left of $z = 1.86$ is 0.9686, so $p = 1 - 0.9686 = 0.031$.
- Alternatively, the **critical value** of z leaving 0.05 in the right tail is **1.645**.
- The tail area beyond the critical value is α (one-tailed) or $\alpha/2$ (two-tailed).

(The textbook rounds σ_p to 0.089 and reports $z = 1.88$, $p = 0.030$ – same conclusion.)

Figure 5. Probability of z with a one-tailed test



Steps 6–8: decide and conclude

Steps 6 and 7. Two equivalent ways to reach the statistical decision:

If probability of TS_{CAL} < significance level α	$\Rightarrow$ reject H_0
If $\lvert TS_{CAL} \rvert > \lvert TS_{CR} \rvert$	$\Rightarrow$ reject H_0

- $p = 0.031 < 0.05$ and $z = 1.86 > 1.645 \Rightarrow$ **reject** H_0.
- Note the sign shift: the larger TS_{CAL}, the smaller its probability under H_0.

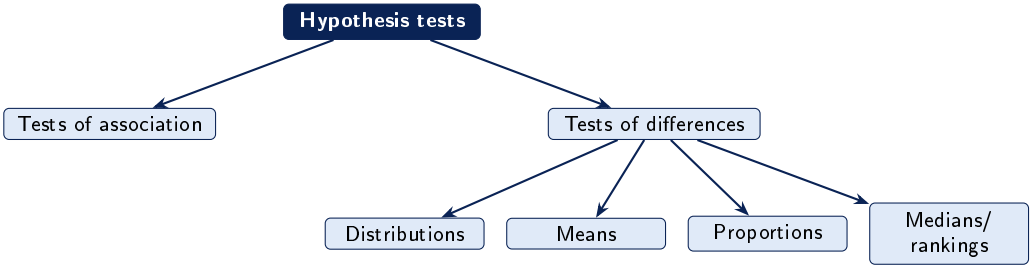
Step 8. Express the decision in terms of the **marketing research problem**:

Conclusion

There is evidence that the proportion of internet users who shop online is significantly greater than 0.40 – the recommendation is to **launch the online store**.

Tests of association vs tests of differences

Figure 6. A broad classification of hypothesis testing procedures



Association: H_0: '... is not related to ...' – examined with cross-tabulations.

Differences: H_0: '... is not different from ...' – examined with z, t and non-parametric tests.

Cross-tabulation

Questions about one variable often lead to questions linking it to others:

- How many brand-loyal users are male?
- Is product use (heavy, medium, light, non-users) related to interest in outdoor activities?
- Is familiarity with a new product related to age and income? Is ownership related to income?

Cross-tabulation (contingency table)

A technique that describes **two or more variables simultaneously**, producing tables that reflect their **joint distribution**. Each variable has a limited number of categories; the table has a cell for every combination of categories.

Why so widely used?

1. Easily interpreted by managers who are not statistically oriented.
2. Clear interpretation links research results to managerial action.
3. A series of cross-tabs may give more insight than one multivariate analysis.
4. May alleviate sparse cells in discrete multivariate analysis; simple to conduct.

Two variables: gender and internet usage

Light users: ≤ 5 hours/week; heavy users: > 5 hours/week.

Table 20.3 Counts

Usage	Male	Female	Row
Light	5	10	15
Heavy	10	5	15
Column	15	15	30

Table 20.4 Column %

Usage	Male	Female
Light	33.3%	66.7%
Heavy	66.7%	33.3%
Column	100%	100%

Table 20.5 Row %

Usage	Male	Female	Row
Light	33.3%	66.7%	100%
Heavy	66.7%	33.3%	100%

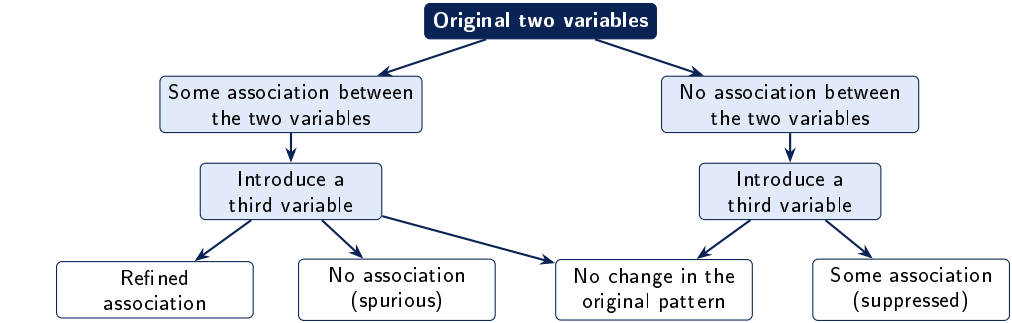
Rule for computing percentages

Compute percentages **in the direction of the independent variable, across the dependent variable**. Gender is independent, usage dependent $\Rightarrow$ Table 20.4 is correct: 66.7% of males vs 33.3% of females are heavy users.

Table 20.5 would imply that heavy usage 'causes' people to be male – implausible. The margins of a cross-tab repeat the one-way frequency tables.

Three variables: introducing a control variable

Figure 7. The introduction of a third variable in cross-tabulation



The following examples use contrived samples of 1,000 participants – a common size in commercial research.

1. Refining an initial relationship

Table 20.6 Purchase of luxury branded clothing by marital status

Purchase	Married	Unmarried
High	31%	52%
Low	69%	48%
Column	100%	100%
n	700	300

Table 20.7 ... by marital status and gender

	Male		Female	
Purchase	Married	Unmarried	Married	Unmarried
High	35%	40%	25%	60%
Low	65%	60%	75%	40%
Column	100%	100%	100%	100%
n	400	120	300	180

- Initially: 52% of unmarried vs 31% of married participants are high purchasers.
- Gender (chosen on the basis of past research) is introduced as a third variable.
- Females: 60% vs 25% – a large gap. Males: 40% vs 35% – much closer.
- **Refined**: unmarried people are more likely to be high purchasers, and the effect is **much more pronounced for females**.

2. Initial relationship was spurious

Table 20.8 Ownership of expensive cars by education

Own	Degree	No degree
Yes	32%	21%
No	68%	79%
Column	100%	100%
n	250	750

Table 20.9 ... by education and income

	Low income		High income	
Own	Degree	No degree	Degree	No degree
Yes	20%	20%	40%	40%
No	80%	80%	60%	60%
Column	100%	100%	100%	100%
n	100	700	150	50

- An agency promoting a car costing more than €80,000 sees that 32% of graduates vs 21% of non-graduates own an expensive car.
- Controlling for **income**, ownership is identical for graduates and non-graduates within each income group.
- The association between education and car ownership **disappears**: it was **spurious** (income drives both).

3. Revealing a suppressed association

Table 20.10 Desire to travel abroad by age

Desire	Under 45	45+
Yes	50%	50%
No	50%	50%
Column	100%	100%
n	500	500

Table 20.11 ... by age and gender

	Male		Female	
Desire	Under 45	45+	Under 45	45+
Yes	60%	40%	35%	65%
No	40%	60%	65%	35%
Column	100%	100%	100%	100%
n	300	300	200	200

- Aggregated, age appears unrelated to desire to travel abroad.
- Men: younger people want to travel more (60% vs 40%). Women: the pattern is **reversed** (35% vs 65%).
- Opposite effects cancel out when data are pooled across gender.
- Controlling for gender reveals the **suppressed association** (a *suppressor effect*).

4. No change in the initial relationship

Table 20.12 Eating frequently in fast-food restaurants by family size

Eat often	Small	Large
Yes	65%	65%
No	35%	35%
Column	100%	100%
n	500	500

Table 20.13 ... by family size and income

	Low income		High income	
Eat often	Small	Large	Small	Large
Yes	65%	65%	65%	65%
No	35%	35%	35%	35%
Column	100%	100%	100%	100%
n	250	250	250	250

- Families were split into small/large (median split); no association is observed.
- Introducing income (median split) still shows no association.
- The third variable **does not influence** the relationship between the first two – **no change**.

General comments on cross-tabulation

- More than three variables can be cross-tabulated, but interpretation becomes **very complex**.
- The number of cells grows **multiplicatively** $\Rightarrow$ hard to keep enough cases per cell.
- Rule of thumb: at least **five expected observations per cell** for the statistics to be reliable.
- Cross-tabulation is therefore inefficient when there are more than a few variables.
- Cross-tabulation examines **association, not causation**. To study causation, use a causal research design (Chapter 11).

Check your understanding (Activities 12.1–12.2)

Define a **spurious association**. What is a **suppressed association**, and how is it revealed?

Significance and strength of association

- **Statistical significance** of the observed association $\rightarrow$ **chi-square** (χ^2).
- **Strength** (degree) of association – important from a practical perspective, and of interest **only if the association is significant**:
  - phi coefficient (ϕ)
  - contingency coefficient (C)
  - Cramer's V
  - lambda coefficient (λ)
  - for ordinal variables: tau b, tau c, gamma

Chi-square statistic

H_0: **no association** between the variables. Compare observed frequencies f_o with the **expected frequencies** f_e that would occur if there were no association:

$$f_e = \frac{n_r \, n_c}{n} \qquad \chi^2 = \sum_{\text{all cells}} \frac{(f_o - f_e)^2}{f_e}$$

where n_r = row total, n_c = column total, n = sample size.

Worked example – Table 20.3 (gender × internet usage)

Every cell: $f_e = \dfrac{15 \times 15}{30} = 7.5$

$$\chi^2 = \frac{(5-7.5)^2}{7.5} + \frac{(10-7.5)^2}{7.5} + \frac{(10-7.5)^2}{7.5} + \frac{(5-7.5)^2}{7.5} = 0.833 \times 4 = 3.333$$

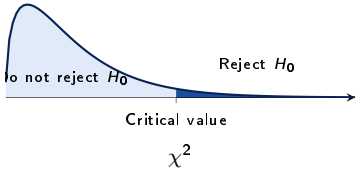
Degrees of freedom: $df = (r-1) \times (c-1) = (2-1) \times (2-1) = 1$

The greater the discrepancy between observed and expected frequencies, the larger χ^2.

The chi-square distribution and decision

- A **skewed** distribution whose shape depends only on df; it becomes more symmetrical as df increases.
- Reject H_0 only if χ^2_{CAL} exceeds the **critical value** for the appropriate df.
- $df = 1$, $\alpha = 0.05$: critical value $= 3.841$.
- $3.333 < 3.841 \Rightarrow$ **do not reject** H_0: the association is not significant at 0.05.
- Mainly due to the small sample: multiply every cell by 10 ($n = 300$) and $\chi^2 = 33.33$ – significant.

Figure 8. Chi-square test of association



Conditions for using χ^2

Use only on **counts** (convert % to numbers) · observations drawn **independently** · not if any **expected** cell frequency < 5 · for a 2×2 table or cells with < 10 observations apply a **correction factor** (here $\chi^2 = 2.133$, still not significant).

Phi, contingency coefficient and Cramer's V

Phi coefficient ϕ

For 2×2 tables only.

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

0 = no association; 1 = perfect association (all cases on one diagonal).

Table 20.3: $\sqrt{3.333/30} = 0.333$

Contingency coeff. C

Any table size.

$$C = \sqrt{\frac{\chi^2}{\chi^2 + n}}$$

Between 0 and 1, but the maximum depends on table size – compare only tables of the **same size**.

Table 20.3: $\sqrt{3.333/33.333} = 0.316$

Cramer's V

Tables larger than 2×2.

$$V = \sqrt{\frac{\chi^2/n}{\min(r-1,\, c-1)}}$$

Ranges from 0 to 1; shows *how strong*, not *how*, variables are associated. $V = \phi$ for 2×2.

Table 20.3: 0.333

In all three cases the association in Table 20.3 is not very strong (being non-significant, it would not normally be computed – shown for illustration).

Lambda and ordinal measures

Lambda coefficient (nominal variables)

- **Asymmetric lambda**: percentage improvement in predicting the dependent variable given the independent variable. From 0 (no improvement) to 1 (prediction without error).
- Computed with each variable in turn as dependent; the two values usually differ.
- **Symmetric lambda**: a kind of average – no assumption about which variable is dependent.
- Table 20.3 (usage dependent): $\lambda = 0.333$ – knowing gender improves prediction by 33%. Symmetric $\lambda = 0.333$.

Ordinal variables – based on concordant, discordant and tied pairs of cases:

Statistic	Adjusts for ties?	Use when
tau b	Yes	Square tables ($r = c$); ranges -1 to $+1$
tau c	Yes	Rectangular tables ($r \neq c$)
gamma	No (nor for table size)	Generally larger values than tau b or tau c

Cross-tabulation in practice

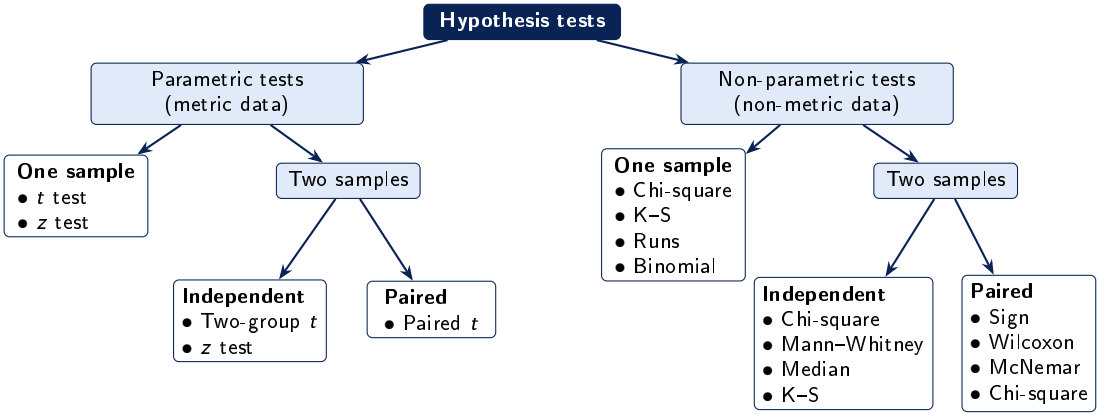
1. Test H_0 of **no association** using χ^2. If you fail to reject H_0, there is no relationship.
2. If H_0 is rejected, determine the **strength** of association (ϕ, C, Cramer's V, λ, ...).
3. If H_0 is rejected, interpret the **pattern** by computing percentages in the direction of the independent variable, across the dependent variable.
4. If variables are ordinal, use tau b, tau c or gamma: magnitude gives strength, sign gives **direction**.
5. Translate significance, strength and pattern into **managerial implications** and recommendations.

Check your understanding (Activity 12.3)

Discuss the reasons for the frequent use of cross-tabulations. What are some of their limitations?

Classification of hypothesis testing procedures

Figure 9. Hypothesis testing procedures



Parametric vs non-parametric; independent vs paired

Parametric tests

Assume the variables are measured on at least an **interval scale** (metric data). Most popular: the *t* test for means.

Non-parametric tests

Assume the variables are measured on a **nominal or ordinal scale** (non-metric data). *Non-examinable in this course.*

Independent samples

Drawn randomly from **different populations**. Data on different groups of participants (e.g. males vs females) are generally treated as independent.

Paired samples

The two sets of observations relate to the **same participants** (e.g. rating two brands, before/after an ad).

The number of samples depends on how data are **treated for analysis**, not how they were collected.

The t test

- Based on Student's **t statistic**: assumes the variable is normally distributed, the mean is known (or assumed), and the population variance is **estimated from the sample**.

$$s_{\bar{X}} = \frac{s}{\sqrt{n}}, \qquad t = \frac{\bar{X} - \mu}{s_{\bar{X}}} \sim t_{n-1}$$

- The **t distribution** is bell-shaped and symmetric like the normal, but with **more area in the tails** (extra uncertainty from estimating σ^2).
- As df increases it approaches the normal; for $n \geq 120$ they are virtually indistinguishable. The t test is quite robust to departures from normality.

Procedure (Activity 12.5)

1. Formulate H_0 and H_1.
2. Select the formula for t.
3. Select α (typically 0.05).
4. Take one or two samples; compute mean and SD.
5. Calculate t assuming H_0 is true.
6. Find df and the p value (or critical t).
7. Reject H_0 if $p < \alpha$ (or $|t_{CAL}| > |t_{CR}|$).
8. Express the conclusion in terms of the marketing research problem.

One sample: t test and z test

Does mean familiarity exceed the neutral value 4.0? $H_0 : \mu \leq 4.0$ $H_1 : \mu > 4.0$ $\alpha = 0.05$

t test (σ estimated from the sample)

$$s_{\bar{X}} = \frac{1.579}{\sqrt{29}} = 0.293$$

$$t = \frac{4.724 - 4.0}{0.293} = 2.471$$

$df = 29 - 1 = 28$; critical $t_{0.05,28} = 1.7011$.
$2.471 > 1.7011 \Rightarrow$ reject H_0.

z test ($\sigma = 1.5$ assumed known)

$$\sigma_{\bar{X}} = \frac{1.5}{\sqrt{29}} = 0.279$$

$$z = \frac{4.724 - 4.0}{0.279} = 2.595$$

Critical z (one-tailed, 0.05) $= 1.645$.
$2.595 > 1.645 \Rightarrow$ reject H_0.

Conclusion

Familiarity with the internet **does exceed** the neutral level of 4.0 (though at 4.724 it is still below 5).

Two independent samples: means

$H_0 : \mu_1 = \mu_2 \qquad H_1 : \mu_1 \neq \mu_2$

If both populations have the **same variance**, pool the sample variances:

$$s^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$s_{\bar{X}_1 - \bar{X}_2} = \sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{s_{\bar{X}_1 - \bar{X}_2}}, \quad df = n_1 + n_2 - 2$$

F test of equal variances

$H_0 : \sigma_1^2 = \sigma_2^2 \qquad H_1 : \sigma_1^2 \neq \sigma_2^2$

$$F_{n_1 - 1,\, n_2 - 1} = \frac{s_1^2}{s_2^2}$$

- $p(F) > \alpha$: do not reject; use **pooled** t.
- $p(F) \leq \alpha$: reject; use **separate-variance** t (approximate, non-integer df).

Examples: users vs non-users' perceptions of a brand; high- vs low-income consumers' leisure spending.

Example: internet usage by gender

Table 20.14 Two-independent-samples t test

	Number of cases	Mean	Std. error mean	F test for equal variances	
				F value	Two-tail p
Male	15	9.333	1.137	15.507	0.000
Female	15	3.867	0.435		

	t value	df	Two-tail p
Equal variances assumed	−4.492	28	0.000
Equal variances not assumed	−4.492	18.014	0.000

- F test: $p < 0.05 \Rightarrow$ variances differ $\Rightarrow$ use 'equal variances **not** assumed'.
- $t = -4.492$, $df = 18.014$, $p = 0.000 < 0.05 \Rightarrow$ reject H_0 of equal means.
- Males (9.333 hrs) use the internet **significantly more** than females (3.867 hrs).

Real research: independent samples t tests

Real research: Shops seek to suit elderly to a 't'

789 participants aged 65+: do **dependent** and **self-reliant** elderly people have different physical requirements for shopping? (1 = not important ... 5 = very important)

Physical requirement	Self-reliant	Dependent	t test p
Delivery to home	1.787	2.000	0.023
Phone-in order	2.030	2.335	0.003
Transportation to store	2.188	3.098	0.000
Convenient parking	4.001	4.095	0.305
Location close to home	3.177	3.325	0.137
Variety of shops close together	3.456	3.681	0.023

Dependent elderly people are more likely to look for home delivery, phone orders, accessible transportation and a variety of shops located close together.

Two independent samples: proportions

Is the proportion who shop online the same for males and females? $H_0 : \pi_1 = \pi_2$ $H_1 : \pi_1 \neq \pi_2$
$\alpha = 0.05$

$$z = \frac{P_1 - P_2}{s_{P_1 - P_2}}$$

$$s_{P_1 - P_2} = \sqrt{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$P = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

Calculation (Table 20.1)

$P_1 - P_2 = 11/15 - 6/15 = 0.733 - 0.400 = 0.333$
$P = (15 \times 0.733 + 15 \times 0.400)/30 = 0.567$
$s_{P_1 - P_2} = \sqrt{0.567 \times 0.433\,(1/15 + 1/15)} = 0.181$
$z = 0.333/0.181 = 1.84$

- Two-tailed: critical value leaves $\alpha/2 = 0.025$ in each tail $\Rightarrow z_{CR} = 1.96$.
- $1.84 < 1.96 \Rightarrow$ **do not reject** H_0.
- The difference (73% vs 40%) is substantial but **not significant** – the samples are small (15 each).

Paired samples t test

Same participants measured twice $\Rightarrow$ form the **difference variable** D_i and test its mean.

$$H_0 : \mu_D = 0 \qquad H_1 : \mu_D \neq 0$$

$$t_{n-1} = \frac{\bar{D} - \mu_D}{s_{\bar{D}}}$$

$$\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n}, \qquad s_D = \sqrt{\frac{\sum_{i=1}^{n} (D_i - \bar{D})^2}{n-1}}$$

$$s_{\bar{D}} = \frac{s_D}{\sqrt{n}}, \qquad df = n - 1$$

Table 20.15 Attitude to internet vs technology

Variable	n	Mean	SD	SE
Internet attitude	30	5.167	1.234	0.225
Technology attitude	30	4.100	1.398	0.255

$\bar{D}$	s_D	SE	Corr.	t	df
1.067	0.828	0.1511	0.809	7.059	29

Two-tail $p = 0.000$

- $t = 1.067/0.1511 = 7.06$, $p < 0.001 \Rightarrow$ reject H_0: attitudes towards the internet are **more favourable** than towards technology.
- Implication: ISPs need not hesitate to target consumers who are not especially tech-savvy.

Real research: paired t tests

Real research: Seconds count

83 media directors of Canadian advertising agencies rated **15-second** and **30-second** TV commercials (1 = excellent ... 5 = poor). Each director rated both lengths – paired samples.

Dimension	15 seconds	30 seconds
Brand awareness	2.5	1.9
Main idea recall	2.7	2.0
Persuasion	3.7	2.1
Ability to tell an emotional story	4.3	1.9

Paired t tests showed all differences were significant: 30-second commercials were rated more favourably on every dimension.

Differences in **proportions** for paired samples use the McNemar or chi-square test (non-parametric).

Non-parametric tests at a glance (non-examinable)

Used when variables are **non-metric**; available for one, two independent and two related samples.

One sample

- **K–S** goodness of fit: $K = \max|A_i - O_i|$
- Chi-square goodness of fit
- Runs test (randomness)
- Binomial test

Two independent

- **Mann–Whitney** U (counterpart of two-group t)
- Median test (less powerful)
- K–S two-sample
- Chi-square (2×2)

Paired

- **Wilcoxon** matched-pairs signed-ranks (counterpart of paired t)
- Sign test (less powerful)
- McNemar (binary)

Internet data: K–S shows usage does not deviate from normality ($K = 0.222 < 0.248$, $p = 0.103$); Mann–Whitney confirms males use the internet more ($z = -3.406$, $p = 0.001$); Wilcoxon confirms attitudes to the internet exceed those to technology ($z = -4.207$, $p = 0.000$).

Summary of hypothesis tests (Table 20.19)

Table 20.19 A summary of hypothesis testing

Sample	Application	Scaling	Test / comments
One sample	Distributions	Non-metric	K–S and chi-square for goodness of fit; runs test; binomial test
One sample	Means	Metric	*t* test if variance unknown; *z* test if variance known
One sample	Proportions	Metric	*z* test
Two independent	Distributions	Non-metric	K–S two-sample test for equivalence of two distributions
Two independent	Means	Metric	Two-group *t* test; *F* test for equality of variances
Two independent	Proportions	Metric / non-metric	*z* test / chi-square test
Two independent	Rankings/medians	Non-metric	Mann–Whitney *U* (more powerful than median test)
Paired	Means	Metric	Paired *t* test
Paired	Proportions	Non-metric	McNemar test for binary variables; chi-square
Paired	Rankings/medians	Non-metric	Wilcoxon matched-pairs signed-ranks (more powerful than sign test)

Check your understanding (Activities 12.4 and 12.6)

Present a classification of hypothesis testing procedures. What is the major difference between parametric and non-parametric tests?

Real research: international brand equity

Real research: The name of the game

100 brand and product managers were surveyed: for international markets, 39% said uniform packaging was *not* a good idea and 38% said it was – packages must fit local linguistic and regulatory needs.

Research question: do consumers in different countries prefer global brand names with packaging customised to local needs?

H_0: standardised branding with customised packaging for a well-established brand will *not* lead to greater market share internationally.
H_1: other factors equal, it *will* lead to greater market share.

Test: take a brand with a mixed strategy (e.g. Colgate); compare market share in countries with standardised vs customised packaging, controlling for other factors, using a **two-independent-samples t test**.

Running the analyses in Python (1/3): data and frequencies

```python
import numpy as np
import pandas as pd
from scipy import stats

# Table 20.1 saved as a CSV file
df = pd.read_csv("internet_usage.csv")
df["familiar"] = df["familiar"].replace(9, np.nan)  # missing

# Frequency distribution (Table 20.2)
x = df["familiar"]
freq = x.value_counts(dropna=False).sort_index()
pct = x.value_counts(normalize=True).sort_index() * 100
print(pd.DataFrame({"N": freq, "valid %": pct.round(1),
                    "cum %": pct.cumsum().round(1)}))

# Location, variability and shape
print(x.agg(["mean", "median", "std", "var", "skew", "kurt"]))
print(x.mode())

# Histogram (Figure 1)
x.plot.hist(bins=range(1, 9), rwidth=0.8)
```

Output (matches the textbook)

mean 4.724 median 5 mode 6
std 1.579 var 2.493
skew −0.094 kurt −1.261

- Recode the missing-value code (9) to `NaN` **before** computing statistics – pandas then excludes it automatically, like 'valid %'.
- `.skew()` and `.kurt()` report *excess* kurtosis (0 for a normal distribution).

Running the analyses in Python (2/3): cross-tabulation

```python
from scipy.stats.contingency import association

# Light (<= 5 hrs/week) vs heavy users
df["grp"] = np.where(df["iusage"] <= 5, "Light", "Heavy")

# Counts (Table 20.3) and column % (Table 20.4)
counts = pd.crosstab(df["grp"], df["gender"])
colpct = pd.crosstab(df["grp"], df["gender"],
                     normalize="columns") * 100

# Chi-square test of association
chi2, p, dof, expected = stats.chi2_contingency(
    counts, correction=False)

# Strength of association
v = association(counts, method="cramer")   # = phi (2x2)
c = association(counts, method="pearson")  # contingency C
```

Output

$\chi^2 = 3.333, \quad df = 1, \quad p = 0.068$
expected $= 7.5$ in every cell
Cramer's $V = \phi = 0.333 \qquad C = 0.316$

- `normalize="columns"` gives percentages in the direction of the independent variable (gender).
- `correction=False` gives Pearson's χ^2; the default (`True`) applies Yates' correction for 2×2 tables ($\chi^2 = 2.133$).
- Check the `expected` array: all cells should be ≥ 5.
- Third variable: `pd.crosstab(df.a, [df.c, df.b])`.

Running the analyses in Python (3/3): tests of differences

```python
from scipy.stats import norm

# One-sample t test: H1 mean familiarity > 4
stats.ttest_1samp(x.dropna(), 4, alternative="greater")

# One-sample z test for a proportion: H1 pi > 0.40
p_hat, pi0, n = 17/30, 0.40, 30
z = (p_hat - pi0) / np.sqrt(pi0 * (1 - pi0) / n)
p_value = norm.sf(z)                    # right-tail area

# Two independent samples (Table 20.14)
m = df.loc[df["gender"] == 1, "iusage"]
f = df.loc[df["gender"] == 2, "iusage"]
stats.levene(m, f, center="mean")       # equal variances?
stats.ttest_ind(m, f, equal_var=False) # separate variances

# Paired samples (Table 20.15)
stats.ttest_rel(df["iattitude"], df["tattitude"])
```

Output

One-sample	$t = 2.470$, $p = 0.010$
Proportion	$z = 1.86$, $p = 0.031$
Levene	$F = 15.507$, $p < 0.001$
Welch	$t = 4.492$, $df = 18.01$
Paired	$t = 7.059$, $p < 0.001$

- `alternative="greater"` or `"less"`: one-tailed test.
- Levene (`center="mean"`) = the F test in Table 20.14.
- Non-parametric: `mannwhitneyu`, `wilcoxon`, `kstest`.

Exercises (1/3)

1. In each situation, which analysis and test statistic would you use?
   a. Preference for The Body Shop shampoo (11-point Likert scale) is measured, consumers see a commercial, and preference is measured again – did the commercial change preferences?
   b. Does preference for The Body Shop shampoo follow a normal distribution?
   c. 1,000 households are classified as heavy, medium, light and non-users of ice cream and as high-, medium- or low-income – is ice cream consumption related to income?
   d. 2,000 households rank 10 supermarkets, including Lidl; the sample is split into small and large households (median split) – does preference for Lidl vary by household size?

Solutions

a. Paired samples t test on the change in mean score.
b. Kolmogorov–Smirnov one-sample test (K = largest absolute difference from the normal).
c. Cross-tabulation, χ^2 test, and Cramer's V for strength.
d. Non-parametric two-independent-samples test: Mann–Whitney, median or K–S.

Exercises (2/3)

2. Red Bull's advertising campaign will be changed if **less than 30%** of consumers like it. A random sample of 300 consumers finds 84 who like it. Formulate the hypotheses, discuss the errors, choose a test and decide.

Solution

$H_0 : \pi = 0.30$ vs $H_1 : \pi < 0.30$ (one-tailed) – z test for a proportion.
Type I error: changing a campaign that at least 30% like. Type II error: keeping a campaign that fewer than 30% like.
$p = 84/300 = 0.28, \quad \sigma_p = \sqrt{0.30 \times 0.70/300} = 0.0265, \quad z = (0.28 - 0.30)/0.0265 = -0.76$
p value $= P(Z \leq -0.76) \approx 0.22 > 0.05 \Rightarrow$ do not reject H_0: **do not change the campaign**.

3. Refrigerators sold in a New Year sale at 10 stores: 80, 110, 0, 40, 70, 80, 100, 50, 80, 30.
Is there evidence that on average **more than 50** were sold per store ($\alpha = 0.05$)? What assumption is necessary?

Solution

One-sample t test: $H_0 : \mu = 50$ vs $H_1 : \mu > 50$. $\quad \bar{x} = 64, \quad s = 33.73, \quad n = 10$
$t = (64 - 50) \,/\, (33.73/\sqrt{10}) = 1.313, \quad p = P(T_9 \geq 1.313) \approx 0.11 > 0.05 \Rightarrow$ do not reject H_0.
No significant evidence of more than 50 per store. Assumption: sales are **normally distributed**.

Exercises (3/3): Benetton pre-test in Python

45 Benetton users: usage (1 = light, 2 = medium, 3 = heavy), gender (1 = female, 2 = male) and 7-point ratings of awareness, attitude, preference, intention and loyalty (9 = missing). Load `Benetton.xlsx` with `pd.read_excel()` and recode 9 to `NaN`.

Question	Python	Result
Usage related to gender?	`chi2_contingency`	$\chi^2 = 6.341$, $df = 2$, $p = 0.042$: males are heavier users (reject at 5%, not 1%; one expected count is $4.67 < 5$)
Mean awareness > 3.0?	`ttest_1samp`	$t = 4.162$, $df = 43$, $p = 0.000$: awareness exceeds 3.0
Gender differences?	`ttest_ind`	Awareness differs ($p = 0.021$; males 4.86 vs females 3.57); attitude ($p = 0.064$) and loyalty ($p = 0.372$) do not
Awareness $>$ loyalty?	`ttest_rel`	$t = 0.62$, $df = 42$, $p = 0.538$: no significant difference
Awareness normal?	`kstest`	$p = 0.268$: normality not rejected
Preference normal?	`kstest`	$p = 0.146$: normality not rejected

Results from the course solutions. `stats.kstest(x, "norm", args=(x.mean(), x.std()))` may give slightly different K–S p values (different approximation); conclusions are unchanged.

Discussion points

Discussion point 1

'Why waste time doing basic forms of data analysis? Why not just go straight to multivariate analyses, whose outputs from most software packages include basic analyses?'

Discussion point 2

Why do managers find cross-tabulations so appealing? What would it take to make managers more appreciative of statistical analyses that go beyond cross-tabulations?

Hints: multivariate techniques have stringent assumptions, are harder to interpret and costlier; basic analysis gives quick, cheap insight and suggests which advanced techniques to use. Cross-tabs present **segment differences** in the simplest way – managers who understand their limits may be drawn to more sophisticated methods.

Summary

- **Frequency distributions** summarise one variable: location, variability and shape.
- **Hypothesis testing** follows eight steps, from H_0/H_1 to a marketing research conclusion; balance Type I and Type II errors.
- **Cross-tabulations** describe joint distributions; χ^2 tests significance and ϕ, C, V, λ measure strength.
- A **third variable** can refine an association, show it was spurious, reveal a suppressed one, or leave it unchanged.
- **Tests of differences**: z / t for one sample, two independent samples (F test first) and paired samples.
- Basic analysis is quick, cheap and easy for managers to understand – and it guides multivariate work.